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INDUSTRIES



FOREX DSS

Enterprise Computing Phase 2

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Stage 1:

1.1. Stakeholder Identification and Analysis

The Forex Trading Decision Support System developed in Phase 1 serves multiple distinct stakeholder groups, each with different requirements for data visualisation and decision making support. Identifying these stakeholders is essential to understanding how data must be presented to meet their specific needs and operational contexts. The primary stakeholder groups for this DSS are novice to intermediate Forex traders, trading coaches and mentors, institutional fund managers, and system administrators. Each group requires fundamentally different visualisations because they operate at different levels of decision making granularity and possess different levels of technical trading knowledge. These four groups are carried Consistently through every stage of this portfolio, and no fifth group is introduced later, so the analysis, the Research and the build all address the same four stakeholders.

1.2. Analysis of Stakeholder Requirements

Novice to Intermediate Traders (Primary End Users)

Novice and intermediate Forex traders represent the primary stakeholders for whom the DSS was originally designed. These users require data visualisations that translate complex technical analysis into clear, actionable trading signals without requiring deep understanding of the underlying mathematical models. Their primary need is immediate signal recognition and rapid decision making under time pressure, since currency markets move continuously during trading sessions. These traders must quickly understand entry price, stop loss level, take profit target, and confidence score without requiring extensive explanation or technical context.

The data story for this stakeholder is straightforward: “take this trade now, or wait.” The trend they must interpret is the immediate direction of the signal, the pattern is whether the current setup resembles past profitable setups, and the relationship is between the confidence score and the risk-reward tier offered. The visualisation purpose for this stakeholder group is to reduce cognitive load by presenting

signal information in a format that requires minimal interpretation, enabling traders to execute recommendations with confidence within seconds of signal generation.

Trading Coaches and Mentors (Educational Stakeholders)

Trading coaches who work with novice traders represent a secondary but highly important stakeholder group. These users require visualisations that expose the reasoning behind each signal so they can teach their mentees why a particular setup is valid and how the system evaluated market conditions.

The data story for this stakeholder is “observe how the system reached this decision.” The trend they interpret is how filter outcomes change across different market conditions, the pattern is which combinations of filters tend to precede profitable trades, and the relationship is between displacement strength and trade outcome. Coaches need visibility into the reasoning pipeline including which structural events triggered, how the displacement ratio (the size of the breakout candle body compared to the recent average body) was calculated, and which market condition filters blocked invalid signals. The visualisation purpose here shifts from rapid decision-making to transparency and education.

Institutional Fund Managers and Risk Officers (Governance Stakeholders)

Institutional fund managers and compliance officers occupy a third stakeholder group with distinct visualisation requirements. These users are concerned with portfolio-level aggregate statistics, cumulative profitability, risk exposure, win-rate consistency, and drawdown management across multiple currency pairs and time horizons. They do not require individual signal-level detail; instead, they require summary dashboards showing whether the DSS is performing within risk tolerance bands.

The data story for institutional stakeholders is “is this system preserving capital and delivering acceptable returns within our risk parameters?” The translation they interpret is the cumulative equity curve over weeks and months, the pattern is the consistency of win rate across pairs, and the relationship is between risk taken (drawdown) and reward earned (return). The visualisation purpose is high-level governance and reporting to compliance frameworks.

System Administrators and Technical Analysts (Operational Stakeholders)

System administrators and technical analysts who monitor DSS health represent a final stakeholder group. These users require visualisations focused on system operability, error rates, data freshness, processing latency, and market data quality. Their data story centres on system reliability, “is the indicator executing correctly on all configured pairs, are real time data feeds maintaining integrity, are there any edge cases or anomalies requiring attention?” This stakeholder requires different visualisation altogether, focused on operational metrics rather than trading outcomes.

1.3. Explanation of How Stakeholders Use DSS Outputs to Support Decision Making

Trader Level Decision Making

Novice and intermediate traders use the DSS output to make a single, time-critical decision on every new bar, namely whether to execute the current setup or stand aside. When the headline cell returns TAKE, the trader reads the six action parameters, being direction, confidence, risk to reward, entry, stop and target, and places the trade within seconds, trusting the confidence score to size the position and the risk to reward tier to set the target. When the headline returns REJECT, the trader waits for the next qualifying bar rather than forcing an entry. The decision is therefore binary and made under time pressure, and the trader dashboard supports it by collapsing the eight gate evaluation into one legible verdict, so the trader acts on the system's judgement rather than re-deriving it. This is the fastest decision cycle of any stakeholder, repeating on every bar during a live session, which is why the trader view is the most heavily reduced of the four.

Coach Level Decision Making

Trading coaches use DSS outputs to make educational decisions about what to teach their students. When reviewing a profitable trade, a coach uses the detailed signal explanation and filter status visualisation to build a teaching narrative, "notice how the market structure shift candle was 2.3 times the average body size, which gave us a confidence score of 0.75 for a 2:1 risk reward ratio , this is why the take profit target was set aggressively rather than conservatively." Conversely, when reviewing a rejected setup, the coach uses the filter rejection explanation to teach students about market condition quality, "here you see the regime filter blocked this signal because the market was in consolidation mode, even though the structural setup looked valid , this is how the system protects you from high probability false signals in ranging conditions." The coach focused visualisation must therefore preserve detailed reasoning information and make filter evaluation criteria visible and interpretable.

Institutional Decision Making

Institutional fund managers use aggregate DSS outputs to make strategic capital allocation decisions. At the end of each week or month, they review cumulative win rate, average win size compared to average loss size, maximum drawdown experienced, Sharpe ratio, and whether the DSS is operating within pre established risk tolerance bands. If the DSS shows a 65% win rate with a 2:1 average risk reward, the manager decides whether to allocate additional capital to the strategy. If a drawdown exceeds 8% of allocated capital, the manager may decide to reduce position sizing or temporarily pause the system. These are portfolio level decisions that occur at weekly or monthly intervals, not in real time like trader decisions. The institutional visualisation must aggregate and summarize hundreds of individual trades into performance metrics that institutional compliance frameworks understand.

Operational Decision Making

System administrators use technical health metrics to make operational decisions about system maintenance and troubleshooting. If the visualisation shows that real time data latency has increased from 50 milliseconds to 2 seconds, the administrator investigates the data feed connection. If error rates on a particular currency pair spike above historical baseline, the administrator checks for data anomalies or temporary market disruptions. If the system stops executing signals altogether, the administrator checks broker connection status and licensing. These visualisations support diagnosis and remediation decisions rather than trading or strategy decisions.

1.4. Analysis of Data Integrity Considerations

Data Ownership and Source Reliability

The Forex DSS combines multiple data sources, and understanding the ownership and reliability of each is essential for stakeholder trust. In the Phase 2 Excel implementation, The historical OHLC price data is imported once from a broker or data provider and stored as a static dataset within the workbook, rather than streamed live. This OHLC data originates from regulated trading venues and is highly reliable because it represents actual executed trades. Ownership of this data belongs to the broker or venue that recorded it, and it is used here under the provider's permitted terms for analysis.

Derived indicator data (ATR, EMA, and the displacement and efficiency measures) is calculated internally by the DSS through formulas applied to the stored OHLC data. Its reliability is high because the calculation logic is deterministic and repeatable, but it remains dependent on the quality of the imported price data. If the imported data contains errors, every derived value inherits those errors.

Because the dataset is static, stakeholders must understand that all insights reflect the market conditions captured at the time of import, not the live market. This is an acceptable and deliberate constraint for a planning and decision-support tool, but it means the data's age must always be communicated clearly to prevent misinterpretation.

Data Validation Processes

The DSS implements validation at multiple stages. At input, the worksheet enforces constraints on what the user may enter, so dates must be valid calendar dates and prices must fall within a sensible range for a currency pair. This prevents nonsensical inputs from corrupting downstream calculations. At calculation, division operations are protected by a small non-zero floor so that an empty or malformed cell cannot produce an invalid result. At decision, the system confirmed that all eight filter conditions have been evaluated before a take-or-reject result is issued, so that no signal is produced from

incomplete data. These three checkpoints, being input validation, a calculation guard and a completeness check before output, together form the validation chain that protects the integrity of every figure a stakeholder reads.

Assessment of Risks of Incorrect or Misleading Interpretation

Overconfidence Bias from High Confidence Scoring

A primary risk is that traders place excessive confidence in high confidence scores. The confidence score reflects displacement strength only (how explosively price moved during the market structure shift candle), not the probability of profit. A high-confidence signal can still fail in unfavourable conditions. To mitigate this, trader visualisations define the confidence score as displacement-strength-based rather than a probability of profit, and coach visualisations reinforce that position sizing must always respect the risk-reward ratio regardless of confidence.

Misinterpretation of Rejection Reasons

A second risk is that traders misinterpret why a signal was rejected and conclude the system is broken. If the regime filter blocks signals during consolidation, an unfamiliar trader might believe the DSS has failed when it is in fact protecting against false signals. To mitigate this, filter rejection is always accompanied by a plain-language explanation stating which condition failed and why, so the user understands that rejection is protective.

Data Freshness and Stale Data Interpretation

A critical risk is the misinterpretation of static data as live performance. Because the Phase 2 system evaluates against a stored historical dataset rather than a live feed, every dashboard must state the date range of the underlying data clearly, for example “data imported for the period ending 15 April 2026.” This prevents a stakeholder from assuming the figures reflect current market conditions. Rather than promising automatic real-time refresh, which Excel cannot provide, the system is honest about its static nature and relies on a clearly labelled manual re-import when newer data is required.

Stage 2:

2.1 Stakeholder Identification and Analysis

[Content copied directly from Phase 1 - Section: Stakeholder Groups]

The four stakeholder groups were identified and analysed in full in Stage 1, namely the novice to intermediate trader, the trading coach, the institutional fund manager and the system administrator. Stage 2 does not restate that analysis but builds directly on it, moving from identifying what each stakeholder needs to researching how comparable enterprises have solved the same problem and planning how each requirement will be met in the Excel dashboards. Every design decision that follows in this stage is traced back to the specific stakeholder requirement it serves, so the research and the plan remain stakeholder-focused throughout.

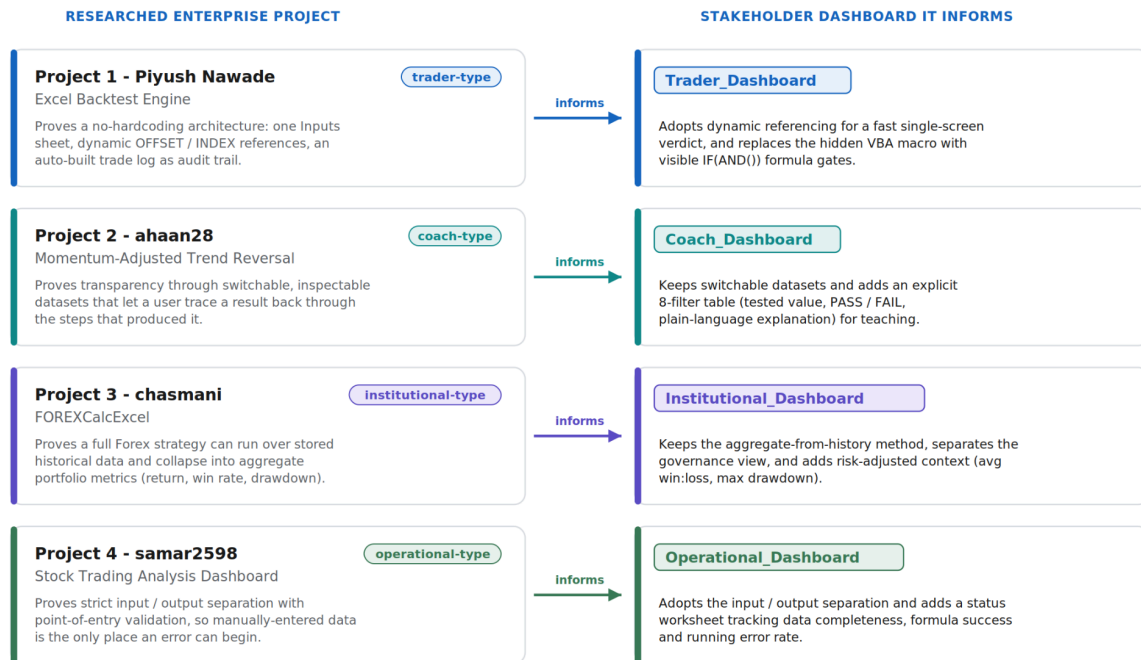
2.2 Research into Similar Trading Dashboards and Financial Visualisation Examples

Because Phase 2 must be delivered entirely in Excel rather than executable Pine Script, research focused on real, published projects that solve the same problem this DSS solves, namely automated signal logic, trade evaluation, and performance reporting built using only spreadsheet formulas and macros instead of a coded indicator. Four projects were selected, each demonstrating an architectural approach that aligns with one of the four stakeholder dashboards.

The diagram below maps each researched project to the stakeholder dashboard it most closely resembles.

Mapping of Researched Enterprise Projects to Stakeholder Dashboards

Each researched Excel project informs the design of one stakeholder dashboard in this DSS



Source: project research and synthesis recorded in Stage 2 (sections 2.2 - 2.4).

Project 1. Piyush Nawade's Excel Backtest Engine (trader-type approach)

Nawade built a fully modular backtesting engine in Excel that generates entry/exit signals from an SMA crossover rule, then auto-logs every trade. The relevant design lesson is that he avoided hardcoding entirely: a single Inputs sheet holds every parameter, and OFFSET and INDEX formulas reference those cells dynamically so changing one value updates the whole workbook. Signal logic lives in a Strategy sheet, with a VBA macro detecting each buy/sell pair and building the trade log automatically. This mirrors the trader dashboard requirement of fast, clean signal output where the user changes inputs and immediately sees the resulting decision without manual recalculation [[An Automated Trading System Built in Excel \(That Actually Tracks Trades, Costs, and Drawdowns\) | by Piyush Nawade | Medium](#)]

The principal strength of this project is the clarity of its single-screen output. Because every parameter lives on one Inputs sheet and every downstream cell references it, the user never has to hunt across the workbook to understand how a result was produced, and the trade log builds itself the instant a new crossover is detected. For a trader stakeholder who must reach a decision quickly, this is precisely the right priority: the cognitive load of interpreting the system is minimised, and the structured trade log doubles as an audit trail that explains what the system did and when.

Its limitation, judged against the needs of this DSS, is that the entire detection step depends on a VBA macro that the user cannot see or interrogate. The crossover logic executes inside code, so a novice

cannot inspect why a particular bar did or did not generate a signal, only that the trade log changed. The single crossover rule is also far simpler than this DSS requires, evaluating one condition rather than eight sequential filters. These limitations directly inform two design decisions. First, this DSS replaces hidden macro detection with visible IF(AND()) formula gates, so the take-or-reject decision can be read and checked in the cells themselves rather than trusted as the output of code. Second, it preserves Nawade's dynamic-referencing discipline, holding all parameters in one place so that the trader dashboard remains as immediately legible as his, while extending the evaluation from one rule to the full eight-filter pipeline.

Project 2. Ahaan28's Momentum-Adjusted Trend Reversal Strategy (coach-type approach)

This GitHub project uses Excel VBA and macros to automate signal generation and build an interactive dashboard that lets the user switch between price data, returns, average returns, and performance charts. Its value as research is the emphasis on transparency: rather than hiding logic, it exposes the intermediate datasets so the user can inspect why the strategy behaved as it did across the backtest period. This is exactly the coach dashboard's purpose, making the reasoning behind each signal visible and switchable for teaching, rather than only showing the final answer [[GitHub - ahaan28/Momentum-Adjusted-Trend-Reversal-Trading-Strategy-using-Excel-VBA-and-Macros: This project highlights how Excel's native automation tools can be leveraged to build practical trading models and backtest their feasibility. The results underline the promise and limitations of simple trend reversal approaches in volatile markets · GitHub](#)]

The principal strength of this project is that it treats the intermediate calculation as something worth showing rather than hiding. By letting the user switch the same dashboard between raw price, period returns, and averaged returns, it lets a viewer trace a result back through the steps that produced it, which is the single most important property a teaching tool can have. A coach using a dashboard built on this principle can move a student from the headline outcome to the underlying numbers without leaving the screen, demonstrating cause and effect rather than asserting it.

Its limitation, against the needs of this DSS, is that the transparency stops at the dataset level rather than the decision level. The user can switch between views of the data, but the project does not present a per-condition pass-or-fail breakdown explaining why a specific signal was accepted or rejected, and its momentum rule again resolves to a single condition rather than a structured set of filters. This informs the coach dashboard design directly: this DSS keeps ahaan28's switchable, inspectable datasets but adds an explicit filter-evaluation table in which each of the eight conditions is shown with its tested value, its pass-or-fail result, and a plain-language explanation. The reasoning is therefore exposed not only at the level of the data, as in ahaan28's project, but at the level of the decision, which is what a coach needs to teach the logic of the system.

Project 3. Chasmani's FOREXCalcExcel (institutional-type approach)

This is the most directly comparable project because it is an Excel-based simulation built specifically for Forex backtesting, the same market this DSS targets. It demonstrates how an entire Forex strategy can be simulated across historical data within Excel and reduced to aggregate performance outputs. This aligns with the institutional dashboard, which needs many individual trades summarised into portfolio-level metrics such as cumulative return, win rate, and drawdown rather than real-time signals [[GitHub - chasmani/FOREXCalcExcel: Excel based simulation for FOREX trading backtesting · GitHub](#)]

The principal strength of this project is its proof that a complete Forex strategy can be run across a body of historical data inside a spreadsheet and collapsed into a small set of summary figures. This is exactly the operation an institutional stakeholder needs, because a fund manager does not read individual trades but asks whether the strategy as a whole preserved capital and produced acceptable returns. By performing the simulation against stored historical data rather than a live feed, it also validates the static-dataset approach this DSS is constrained to use in Excel, confirming that the constraint does not prevent meaningful aggregate analysis.

Its limitation is that it presents a single, undifferentiated set of outputs with no stakeholder-specific framing and no risk-adjusted context. A raw cumulative return figure does not tell a fund manager whether that return was earned safely, and the project offers no equivalent of a Sharpe ratio or a drawdown band against which performance can be judged. This informs the institutional dashboard design in two ways. First, this DSS keeps chasmani's aggregate-from-historical-data method but separates the institutional view from the trader and coach views, so each stakeholder sees only the level of detail relevant to their decision. Second, it adds risk-adjusted and comparative context, presenting win rate, average win against average loss, maximum drawdown and a risk-adjusted return measure alongside the headline figure, so that the aggregate output supports a genuine governance judgement rather than a bare profit number.

Project 4. Samar2598's Stock Trading Analysis Dashboard (operational-type approach)

This MS-Excel dashboard requires the user to enter only a small set of input fields (trade number, stock name, market cap, entry price, stop loss, win/loss flag), and every other column is then calculated automatically from those inputs. Its lesson is the strict separation of manually-entered source data from automatically-derived fields, with validation built into what the user is allowed to enter. This aligns with the operational dashboard, which depends on clean input integrity and automatic status calculation to monitor whether the system is functioning correctly [[GitHub - samar2598/Stock-Trading-Analysis-Dashboard: This is a dashboard created using MS-Excel to analyse the stock related tasks\(eg. trading, back testing strategies etc\) and create visual and interactive interface. · GitHub](#)]

The principal strength of this project is the discipline of its input boundary. By restricting the user to a handful of source fields and deriving everything else by formula, it guarantees that the

manually-entered data is the only place an error can originate, and it makes that data easy to validate at the point of entry. This is the foundation of data integrity in any spreadsheet system, because once the source fields are trustworthy, every derived figure inherits that reliability. For an operational stakeholder whose entire concern is whether the system can be trusted, this clean separation is the most valuable property the project demonstrates.

Its limitation is that it monitors the trade data but not the health of the system that processes it. The dashboard validates what the user types, yet offers no indication of whether the underlying dataset is complete, whether the last calculation cycle succeeded, or how often inputs have produced errors. This informs the operational dashboard design directly. This DSS adopts samar2598's strict input and output separation and point-of-entry validation, then extends it with a dedicated status worksheet that reports the completeness of the historical dataset, the success of the most recent evaluation cycle, and the running error rate. The operational view therefore monitors not only the integrity of the data, as samar2598's project does, but the integrity of the evaluation process itself, which is what a system administrator must be able to confirm before trusting the results.

2.3 Analysis of Strengths and Limitations

Strengths identified across the four projects

All four projects confirm that genuine decision-support logic can be reconstructed in a spreadsheet without code execution. Nawade's dynamic referencing proves that a no-hardcoding architecture keeps a workbook flexible. Ahaan28's switchable datasets prove that filter reasoning can be made transparent for an educational user. FOREXCalcExcel proves the approach scales to Forex specifically and to aggregate reporting. Samar2598's input-then-auto-calculate model proves that strict input/output separation with validation keeps data reliable. Together they validate the core Phase 2 design of eight filter columns resolving to TRUE/FALSE, gated by an IF(AND()) formula, feeding a take-or-reject output.

Limitations identified across the four projects

The same four projects share limitations this DSS must design around. None integrates a live data feed, as every one relies on manual data entry or periodic import, so signals are evaluated on a delay rather than in real time. Their charts are static images that cannot be clicked to drill down. VBA-dependent projects require macro-enabled files and programming knowledge that an end user may not have, which is a reason this DSS favours native formulas over heavy VBA where possible. Larger trade histories also slow recalculation. The Phase 2 design responds by adopting manual update cycles, native-formula logic gates, and carefully chosen summary tables rather than attempting to replicate real-time interactivity.

2.4 Synthesis of Research Findings

The research into the four comparable Excel-based DSS projects produces a consistent set of design principles that directly inform the planning of this DSS. Each project independently arrives at a three-tier architecture separating raw input data, calculated logic, and visualised output across distinct worksheets, confirming that this separation is the established convention rather than an arbitrary choice.

The planned architecture for Excel must account for a critical constraint: Excel cannot integrate live data feeds from brokers. Instead, the architecture stores historical OHLC data in a hidden read-only worksheet. The user manually imports prior bars of EUR/USD data into this hidden sheet, and when they manually enter a new bar's values into the Input worksheet, formulas evaluate that new bar against the historical dataset to determine filter pass or fail status. This manual-trigger evaluation cycle replaces the real-time autonomous execution that Phase 1 Pine Script provided.

The table below maps each limitation found in the researched projects to the specific design decision it informs in this DSS, following the synthesis format used in Phase 1.

Research Finding (Limitation in existing project)	Application to This DSS (Solution)
Nawade's engine relies on a coded VBA macro to detect signals, requiring programming literacy.	This DSS replaces macro detection with native IF(AND()) formula gates so the take-or-reject decision is readable without code.
ahaan28's dashboard exposes returns but does not explain why each filter passed or failed.	The coach dashboard adds a plain-language explanation column for every filter, making the reasoning fully transparent for teaching.
FOREXCalcExcel aggregates results but offers no stakeholder-specific view of the data.	This DSS separates aggregate institutional reporting from the trader and coach views so each stakeholder sees only what they need.
samar2598's dashboard has no system-health or data-integrity monitoring.	The operational dashboard adds a status worksheet tracking data completeness, query success and error rate.
None of the four researched projects integrates a live data feed.	This DSS is designed around a static historical dataset and a manual trigger cycle, matching the genuine constraints of Excel.

2.5 Explanation of UX/UI Design Principles Applied to the Excel Implementation

The research into existing dashboards reveals five user-experience principles that govern effective spreadsheet design, each applied deliberately across the four DSS dashboards. These five principles are not invented for this project but drawn from established interface and dashboard design research. Visibility of system status, consistency and standards, and aesthetic and minimalist design are three of Jakob Nielsen's ten usability heuristics, and they underpin the Feedback, Consistency and Minimalism principles below. Progressive disclosure is a recognised information-design technique for managing complexity without overwhelming the user, and the emphasis on visual hierarchy and on removing non-data ink follows Stephen Few's guidance on information dashboard design and Edward Tufte's data-ink ratio. Each principle is therefore applied as a researched convention rather than a personal preference.

Progressive Disclosure

Progressive disclosure recommends presenting essential information immediately while keeping detailed data accessible on demand. Each dashboard's primary worksheet shows only the single most important output, and granular data is relocated to secondary worksheets reached through tabs and hyperlinks.

This principle is applied differently for each stakeholder according to how much detail their decision requires. The trader sees only the take-or-reject signal and a short recent-trades table, with the full history one tab away, because their decision is immediate and detail would slow it. The coach sees the filter table on the primary view but the underlying formula breakdown only on a linked sheet, so a lesson can begin with the result and descend into the calculator when the moment calls for it. The institutional view shows headline metrics first and the trade-by-trade record last, and the operational view shows status indicators first and detailed logs on demand. In every case the most decision-relevant figure is never hidden, and the supporting depth is always one deliberate step away rather than absent.

Consistency

Consistency recommends that visual language and interaction patterns remain uniform so users develop transferable mental models. Column order, table structure, and the green and red colour convention are standardised across every worksheet so a user who learns one dashboard can navigate the others.

The value of this for a multi-stakeholder system is that knowledge transfers between roles. A coach who also trades, or a fund manager who reviews a coach's screen, meets the same colour meanings, the same column order, and the same table layout on every dashboard, so no relearning is required when

moving between them. The same date format, the same green-for-positive convention, and the same placement of the headline figure at the top-left recur everywhere, which turns the four dashboards into one coherent system rather than four separate tools that happen to share data.

Visual Hierarchy

Visual hierarchy recommends using font size, weight, and spacing to direct attention to the most important information first. The headline output of each dashboard appears in the largest, boldest type at the top of the worksheet, with supporting detail in progressively smaller type below.

Hierarchy is tuned to each stakeholder's single most important question. On the trader dashboard the largest element is the direction and price of the signal, because that is what a decision turns on. On the coach dashboard the filter table is given the central visual weight, because the teaching happens there. On the institutional dashboard the cumulative return occupies the largest type, because that is the first governance question. On the operational dashboard the status indicators are the dominant element, because an administrator needs to confirm soundness before reading anything else. The eye is therefore guided to the right starting point on each screen rather than to a generic one.

Minimalism

Minimalism recommends removing any column, row, or decorative element that does not convey meaning. Each primary view is limited to the essential fields and defers derived metrics to detail worksheets, following samar2598's model.

Minimalism is most important on the trader and operational dashboards, where speed of reading is the whole point, so these views carry the fewest possible elements and no decorative formatting that does not encode data. It is applied more moderately on the coach dashboard, where additional columns such as the explanation text genuinely carry meaning and so are retained. The test applied throughout is whether an element changes a decision or a lesson, and if it does not, it is moved to a detail sheet or removed, which keeps each primary view free of the clutter that would otherwise compete with the information that matters.

Feedback

Feedback recommends that the system communicate the result of a user action immediately. Data validation messages and conditional formatting respond the instant data is entered, recolouring a profit or loss cell green or red without further action.

Because the system runs on a manual trigger rather than a live feed, immediate feedback is what confirms to the user that their action has taken effect. When a trader enters a new bar, the signal cell recolours and the filter results update at once, confirming the evaluation has run. When an invalid value is typed, a validation message appears at the point of entry rather than allowing a silent error to

surface later in a calculation. This responsiveness is the substitute, in a static-data system, for the reassurance that a live system gives through constant updating, and it is the reason the user can trust that what they see reflects what they last entered.

2.6 Description of Input Datasets and Data Architecture for Each Dashboard

The DSS architecture in Excel requires a hidden historical data worksheet and three visible worksheets per stakeholder. All operate on a manual trigger cycle where the user enters a new bar's data and the system evaluates it against the stored historical dataset.

Hidden Historical Data Worksheet

A hidden, read-only worksheet stores 150 bars of prior EUR/USD OHLC data imported from a CSV file. This dataset remains static unless the user explicitly refreshes it. Formulas reference this historical data to calculate baseline metrics such as prior range, recent range, average body size, and volatility bands. This hidden sheet is the source of truth for all comparative calculations, allowing the system to evaluate new bars against established market conditions rather than against real-time data.

Trader Dashboard Data Flow

The trader dashboard draws on three active worksheets. An Input worksheet receives the manually entered new bar data (date, time, open, high, low, close). Upon entry, a chain of native worksheet formulas evaluates all eight filter conditions against the historical dataset. A Logic worksheet calculates the confidence score based on displacement strength and resolves the IF(AND()) gate to produce a take or reject decision. A Summary worksheet displays the current signal in large text along with a table of recent trades, all updated after the trigger evaluation completes.

Coach Dashboard Data Flow

The coach dashboard adds a Signal Analysis worksheet that displays the current bar's filter evaluation in full. Formulas populate the columns showing each filter's condition, the calculated values from the historical data, the pass/fail result, and a plain-language explanation. A secondary Calculation Breakdown worksheet exposes the formula logic so coaches can show students exactly how displacement ratio was calculated from the historical dataset and how that ratio fed into the confidence score.

Institutional Dashboard Data Flow

The institutional dashboard draws from a complete Trade History worksheet recording every evaluation made by the user (manually entered bar plus resulting signal). SUMIF and COUNTIF formulas aggregate this history to calculate cumulative returns, win rates, and monthly performance. A pivot table groups

trades by symbol and calculates metrics per pair. All institutional metrics are calculated from the accumulating trade history stored in the workbook, not from live market data.

Operational Dashboard Data Flow

The operational dashboard requires a single Status worksheet where system administrators manually log or import status checks. A row for Data Feed completeness shows whether the historical dataset has the required 100+ bars. A row for Formula Validation logs whether every evaluation formula resolved without error on the last trigger cycle. A row for Error Rate tracks how many signal evaluations failed due to missing data or formula errors. All status information is manually entered or populated by error-handling formulas after each evaluation cycle.

2.7 Justification of Chart and Visualisation Selections

Trader Dashboard Visualisations

A column chart is selected to display the profit and loss of recent trades because it communicates the ratio of winning to losing trades at a glance, with profits rising green above the zero line and losses falling red below it. A line chart is selected for the cumulative equity curve because it reveals at once whether the account is in drawdown or recovery, which a table of values cannot convey as immediately.

A column chart is preferred over a pie chart for the recent-trade results because a pie chart would show only the proportion of wins to losses and would lose the sequence and magnitude of each trade, both of which a trader needs to read a streak. A line chart is preferred over a column chart for the equity curve because the continuous line emphasises the direction and slope of cumulative performance, whereas separate columns would invite the eye to compare individual heights rather than follow the overall trajectory, which is the very thing a trader is trying to judge.

Coach Dashboard Visualisations

A colour-coded filter table is selected as the primary visualisation because the coach's data story depends on tracing pass and fail logic filter by filter, which a chart cannot express. A scatter plot of displacement ratio against trade outcome is selected to demonstrate the relationship between setup strength and profitability.

A table is preferred over any chart for the filter evaluation because the information is categorical and per-condition: each of the eight filters has a name, a tested value, a result and an explanation, and only a table can hold those four fields side by side for every filter at once. A scatter plot is preferred over a line or bar chart for the displacement-against-outcome relationship because the data is a cloud of independent trades rather than an ordered series, and a scatter plot is the only common chart type that

shows the relationship between two variables across many unconnected points and lets the coach show that the relationship is a tendency rather than a guarantee.

Institutional Dashboard Visualisations

Key performance indicators are selected as large numeric values rather than charts because institutional stakeholders require headline metrics readable in seconds. A line chart of cumulative return reveals the performance trend, a horizontal bar chart of return by symbol exposes which pairs perform well, and a pivot table permits sorting and filtering across the full trade population.

Large numeric indicators are preferred over charts for the headline figures because a single decisive number such as total return or maximum drawdown is read faster as text than as a shape, and precision matters at the governance level. A horizontal bar chart is preferred over a vertical one for the by-symbol comparison because the currency-pair labels are text that sits more legibly at the left of horizontal bars than beneath crowded vertical columns, and the ranking from best to worst pair is read naturally down the page. A pivot table is preferred over a fixed table for the trade population because it lets the manager re-sort and re-group the same data to answer questions as they arise, without the dashboard having to anticipate every query in advance.

Operational Dashboard Visualisations

Traffic-light status indicators are selected because operational monitoring requires an instant judgement of whether all systems are sound, and sparklines are selected to embed a trend within each status row so an administrator can distinguish a chronic problem from an intermittent one.

A traffic-light indicator is preferred over a numeric readout for status because an administrator scanning for problems needs a categorical operational, degraded or failed judgement rather than a raw figure to interpret, and colour with a label delivers that judgement instantly. A sparkline is preferred over a full chart for the trend because it must fit inside a single status row beside the current value, because a full-sized chart would break the one-row-per-component layout that makes the dashboard scannable, whereas the sparkline conveys just enough shape to separate a steady problem from an intermittent spike without leaving the row.

2.8 Justification of Colour Schemes

Colour is justified as a deliberate design decision rather than decoration, because every coloured state in these dashboards is intended to carry meaning that supports a stakeholder decision rather than to ornament the view.. Across every dashboard the system uses a single consistent convention: green denotes a positive or passing state (a profitable trade, a filter that passed, a long signal), red denotes a

negative or failing state (a losing trade, a filter that failed, a short signal), and a neutral grey denotes an inactive or not-applicable state. This convention is chosen because it matches the near-universal financial-market association of green with gain and red with loss, so traders interpret it without instruction.

The colour scheme is applied through conditional formatting so that meaning is generated by the data itself rather than entered manually, which removes the risk of a cell being mislabelled. On the trader dashboard the headline signal cell is shaded green or red by direction to support a decision in seconds. On the coach dashboard the filter status column is shaded per row so a student can see at a glance how many conditions passed. On the institutional dashboard cells are shaded against tolerance bands, so a drawdown beyond the agreed limit turns red as a warning. On the operational dashboard the green, amber and red traffic-light convention signals operational, degraded and failed states.

Because colour alone disadvantages users with colour vision deficiency, every coloured state is paired with a text label or symbol, so the scheme communicates through both hue and word. This dual encoding is the reason the colour decision is justified rather than merely stylistic, and it is examined further under accessibility in section 2.11.

2.9 Justification of Layout and Placement on Dashboards

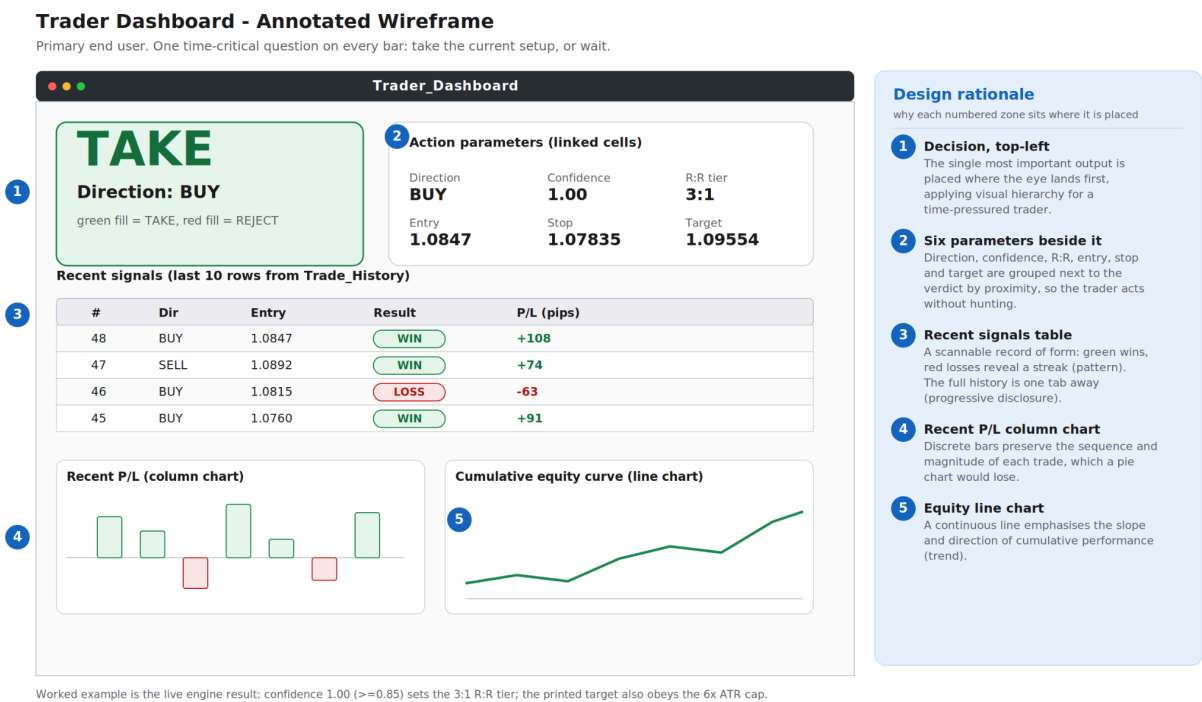
Layout and placement are justified against the reading behaviour of each stakeholder, because the position of an element on a dashboard determines the order in which a stakeholder reads it and therefore how quickly the correct decision is reached. The governing principle is that the eye reads a worksheet from the top-left, so the single most important output for each stakeholder is placed in the top-left region where it is seen first.

On the trader dashboard the current take-or-reject signal is placed at the very top in the largest type, with the recent-trades table beneath it and the charts below that, so the decision is reached before any supporting detail competes for attention. On the coach dashboard the price context sits at the top, the eight-filter evaluation table occupies the central focus where teaching attention is directed, and the resulting trade outcome sits beneath it, so the layout physically follows the teaching narrative from observation to reasoning to result. On the institutional dashboard the headline performance indicators are placed across the top as a summary band, the period-comparison table sits immediately below for context, and the granular trade history is placed last because it is consulted only on demand. On the operational dashboard the status indicators are arranged as a single scannable column so that one downward glance confirms whether every component is sound.

Placement also separates input from output. The hidden historical data and the input-entry area are kept away from the visual output region so that a user cannot accidentally overtyping a calculated cell, which protects data integrity and follows the strict input/output separation observed in the samar2598 project.

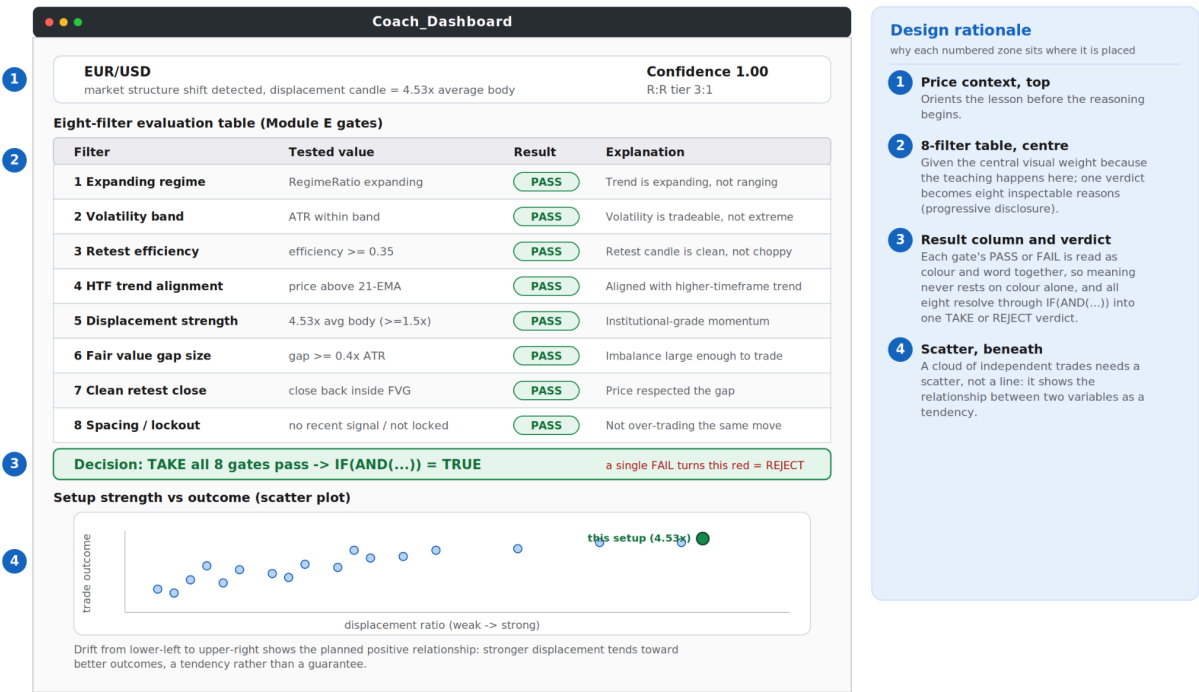
WireFrames

The planned layout for each dashboard is set out in the annotated wireframes below. Each wireframe fixes the position of every zone before the build and maps that position to the UX principle it serves, so the placement decisions described in this section are shown as well as stated. The numbered annotations on each wireframe correspond to the design rationale panel beside it.



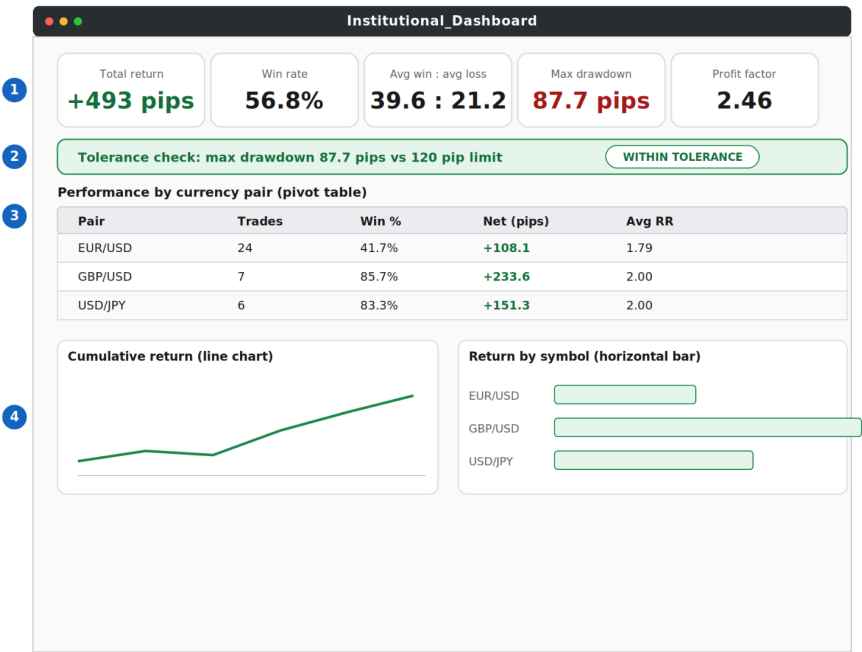
Coach Dashboard - Annotated Wireframe

Educational stakeholder. Decomposes one TAKE / REJECT verdict into eight teachable reasons.



The Coach dashboard wireframe gives the eight filter evaluation table the central position because the coach’s whole task is tracing pass and fail logic filter by filter, which decomposes one verdict into eight teachable reasons and applies progressive disclosure. The result column is placed at the right of each row where colour and text are read together, and the relationship scatter is positioned beneath the table so the lesson moves from individual gates to the overall tendency between setup strength and outcome.

Institutional Dashboard - Annotated Wireframe
Governance stakeholder. Portfolio-level question: is the strategy preserving capital within risk tolerance?

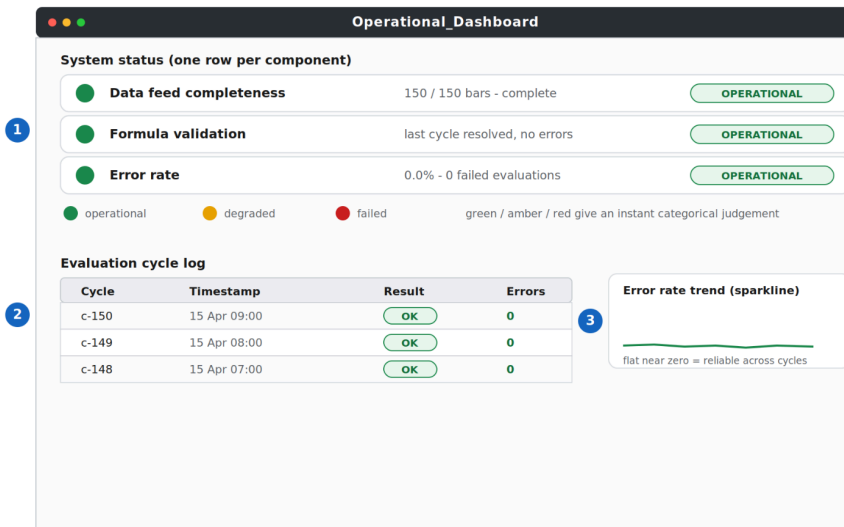


- Design rationale**
why each numbered zone sits where it is placed
- 1 KPI cards, top**
Large numeric values, not charts: a governance reader needs headline metrics readable in seconds, and precision matters.
 - 2 Tolerance flag beneath**
Converts a raw drawdown figure into a within-tolerance / breach judgement at the point of reading.
 - 3 Per-pair pivot table**
A grouped breakdown the manager can re-sort to decide which pairs warrant more or less capital.
 - 4 Line + horizontal bar**
The trend and the categorical comparison sit at the foot so they support, not crowd, the headline figures.

The Institutional dashboard wireframe leads with two rows of key performance indicator cards because a governance reader requires headline metrics readable in seconds, and the tolerance flag sits directly beneath them so a raw drawdown figure is converted into a within tolerance judgement at the point of reading. The per pair table provides the grouped breakdown, and the cumulative line and horizontal bar charts occupy the foot so the trend and the categorical comparison support rather than crowd the headline figures.

Operational Dashboard - Annotated Wireframe

System administrator. Question: is the system sound, so the results can be trusted?



All-green status confirms the stored dataset and the evaluation process are sound (the data is static, so this is integrity, not feed liveness).

Design rationale

why each numbered zone sits where it is placed

- 1 Status table + traffic lights**
An administrator scanning for problems needs a categorical operational / degraded / failed judgement, not a raw figure to interpret.
- 2 Evaluation cycle log beneath**
Holds the data behind the error rate, available on demand (progressive disclosure).
- 3 Compact error sparkline**
Fits inside a single status row, so it shows reliability over time without breaking the one-row-per-component layout.

The Operational dashboard wireframe places the system status table at the top with a traffic light in each row so an administrator scanning for problems reaches an instant operational, degraded or failed judgement. The evaluation cycle log sits below to hold the data behind the error rate, and the compact error trend is positioned beside it so reliability over time is visible without breaking the one row per component layout that keeps the dashboard scannable.

2.10 Explanation of Trends, Patterns, Relationships and Insights Each Dashboard Reveals

Trader Dashboard Insights

The trader dashboard reveals signal consistency and drawdown recovery. Observing that recent comparable signals were profitable builds confidence in executing the current decision, while the equity curve reveals whether recent trades are recovering or extending losses.

Stated precisely, the trend the trader reads is the slope of the equity curve over recent trades, the pattern is the run of greens and reds in the recent-results column that signals a winning or losing streak, and the relationship is between the confidence score attached to a signal and the size of the risk-reward target the system offers for it. The insight that emerges from these together is whether the present moment is a favourable one to act on a signal: a rising curve, a recent run of wins, and a

high-confidence signal align into a clear decision to take the trade, whereas a falling curve and a low-confidence signal counsel caution even when a signal is present.

Coach Dashboard Insights

The coach dashboard reveals filter-logic transparency and the relationship between setup quality and outcome. The filter table provides visual proof of why a setup was accepted or rejected, and the scatter plot reveals that stronger displacement tends toward profitability, supporting the teaching that setup quality matters.

Stated precisely, the pattern the coach draws out is which combinations of passing filters tend to precede profitable trades, the relationship is the upward tendency of profit as displacement strength increases, visible as a drift of points from lower-left to upper-right in the scatter plot, and the trend across many examples is that setups passing all eight filters succeed more often than those scraping through on the minimum. The insight is pedagogical: a student learns that the system's value lies not in any single filter but in the confluence of all of them, and that the strength of a setup, not merely its existence, is what makes it worth taking.

Institutional Dashboard Insights

The institutional dashboard reveals overall profitability and risk, period-comparison trends, and symbol-level variation. Comparing one period against another reveals whether recent performance is typical, and the symbol breakdown reveals which pairs warrant increased or reduced allocation. Stated precisely, the trend is the direction of cumulative return across weeks and months, the pattern is the relative consistency of win rate from one currency pair to the next, and the relationship is between risk taken, read as maximum drawdown, and reward earned, read as return, which together express whether profit was won safely or through excessive exposure. The insight that supports the governance decision is whether the strategy deserves continued or increased capital: a steady upward trend, consistent win rates, and a drawdown within tolerance justify allocation, while strong returns accompanied by a deep drawdown flag a strategy that is profitable but risky and may warrant reduced sizing.

Operational Dashboard Insights

The operational dashboard reveals system soundness and trend information. All-green status confirms results reflect true strategy performance, while a rising error rate across successive evaluation cycles reveals a developing problem. Because the data is static, these insights describe the integrity of the stored dataset and the evaluation process, not the liveness of a feed.

2.11 Description of Accessibility and Inclusive Design Considerations

Colour Reliance

Because the dashboards depend on a green and red convention that disadvantages users with red-green colour blindness, every colour-coded cell is supplemented with a text label, so PASS and FAIL, or WIN and LOSS, are conveyed by word as well as by colour. Charts carry data labels so meaning never rests on colour alone. This dual encoding follows the Web Content Accessibility Guidelines principle that information must never be conveyed by colour alone.

Visual Accessibility

Labels use eleven-point or larger fonts and key indicators use eighteen-point or larger, and the worksheets remain legible under application zoom, with charts resizing to match. These sizes are chosen so that the headline figures remain perceivable at reasonable magnification for users with low vision.

Motor and Non-Mouse Access

The dashboards are fully keyboard-navigable, requiring no drag-and-drop, so users with motor impairments can move between cells, follow hyperlinks, and read cell comments without a mouse.

Screen Reader Compatibility

Critical information is presented in compact summary tables rather than long rows so that screen readers such as JAWS and NVDA can convey the headline data without traversing hundreds of cells, with the institutional summary placed at the top of its worksheet for this reason.

Stage 3:

This stage produces the data visualisation solution in Excel and records, explains and analyses how each stakeholder dashboard was implemented. The workbook builds the four dashboards planned in Stage 2 on top of the existing Forex Decision Support System, with every figure derived from one shared dataset through a transparent formula chain. The development process, the implementation techniques,

the automation of the evaluation cycle and the way each dashboard communicates its trends, patterns and insights are documented section by section, and the actions taken to maintain data integrity and security are set out in full. The live build and evaluation are demonstrated end to end in the accompanying screen recording.

3.1 Description of the Implementation Approach and Workbook Architecture

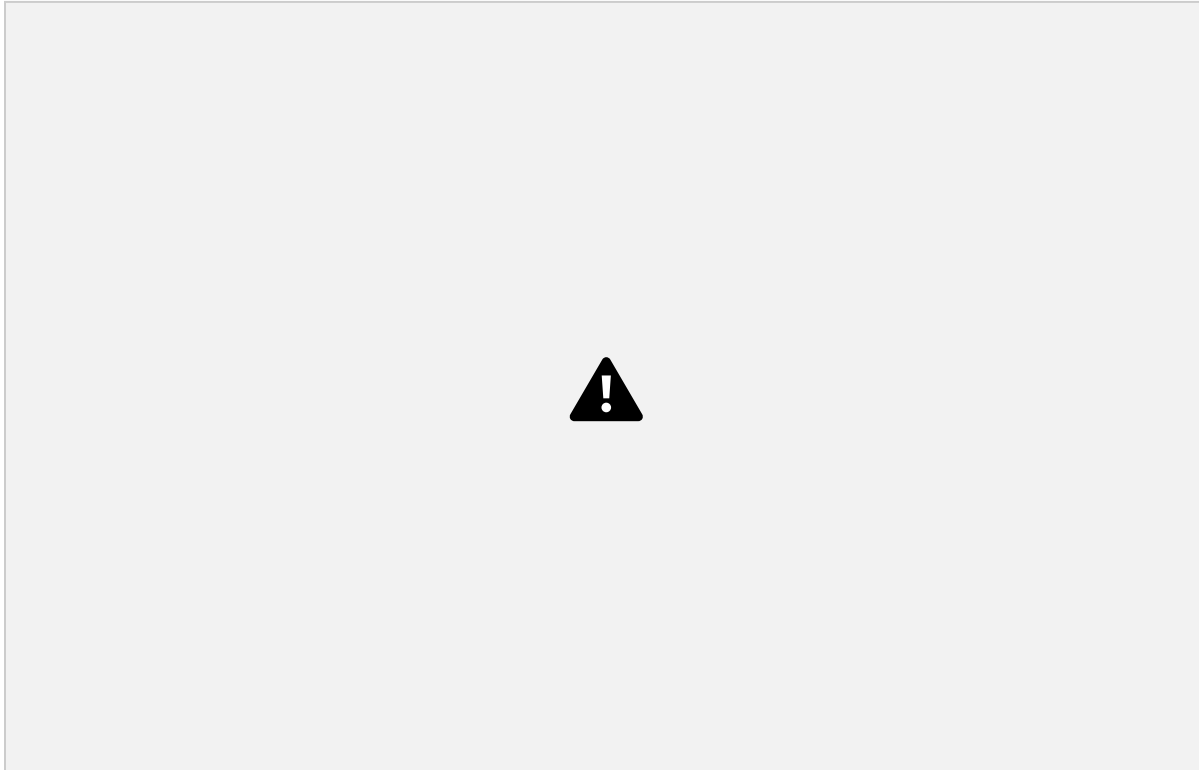
The solution is implemented as a single Excel workbook of ten worksheets that deliberately separates the source data, the manual input point, the evaluation engine and the four stakeholder dashboards into distinct layers. This separation keeps one authoritative dataset feeding every view, which is the consistency principle established in Stage 2 and the single most important architectural decision in the build.

The worksheets fall into four layers. The data layer is the Historical_Data sheet, a static set of 150 EUR/USD bars that acts as the single source of truth. The control layer is the Input sheet, the only place a user types, together with the Logic sheet, which holds the eight filter gates and the displacement, confidence and risk calculations. The communication layer is the four dashboards, being Trader_Dashboard, Coach_Dashboard, Institutional_Dashboard and Operational_Dashboard. A supporting layer holds Calculation_Breakdown, a teaching view that exposes the derivation step by step, and Trade_History, the accumulating log that drives the portfolio aggregates. An Index sheet sits in front of these as a navigation and colour-convention guide. The tabs are ordered so a viewer meets the dashboards first and the engineering behind them second, which carries the progressive disclosure principle from the plan into the structure of the file itself.

Because the layers are connected by cell references rather than copied values, a change entered once on the Input sheet flows through the Logic engine and surfaces simultaneously across every dependent dashboard. No figure is ever typed twice and no dashboard holds a private copy of the data, which removes the most common source of inconsistency in spreadsheet reporting and guarantees that the four stakeholder views can never disagree with one another. The completed workbook contains roughly 1790 formulas and resolves with zero formula errors, confirming that the architecture holds together end to end.

3.2 Description of the Historical Dataset Construction

The Historical_Data sheet was built first because every downstream calculation depends on it. The raw open, high, low and close values for 150 EUR/USD hourly bars covering the period ending 13 March 2026 were exported from the trading platform and imported as the source of truth, after which a set of analytics columns was layered on top to supply the inputs the Decision Support System requires. The export step is shown below.



Every analytics column is produced by an Excel formula rather than a typed number, so the dataset recalculates in full if the underlying prices are ever replaced. Body is the absolute difference between open and close, Range is high minus low, and Efficiency is body divided by range. True Range and a fourteen period ATR are computed for the volatility and risk rules. The twenty one period EMA is built with a recursive formula that weights each close at two divided by twenty two and carries the previous EMA forward, which reproduces the higher timeframe trend reference the system relies on. AvgBody14, RecentRange5, PriorRange5 and a RegimeRatio complete the set so the eight filters have every value they need.

The sheet is hidden and treated as read only. Hiding it keeps the source of truth out of reach of accidental edits while leaving it fully available to the formulas that reference it, which is the first of several data integrity measures described in section 3.9.

3.3 Explanation of the Evaluation Engine Implementation

The Logic sheet is the engine that converts a new bar into a decision, and it was implemented to reproduce the exact rules of the Phase 1 system so that the dashboards report the real behaviour of the Decision Support System rather than a simplified imitation.

The engine first computes the displacement ratio as the candle body divided by the fourteen bar average body, then converts it to a confidence score using the rule that confidence equals the displacement ratio minus one, divided by two, held between zero and one. The confidence score selects the risk to reward tier, where a score at or above 0.85 targets three to one, a score at or above 0.5 targets two to one, and anything lower targets one to one. The eight Module E filters then run as binary gates, each implemented with an IF function that shows its tested value beside a PASS or FAIL result. The gates cover the expanding regime check, the volatility band, retest efficiency, higher timeframe trend alignment, displacement strength, fair value gap size, a clean retest close, and the spacing and lockout check. The final decision is a single IF wrapped around an AND of all eight gates, which returns TAKE only when every gate passes and REJECT otherwise, faithful to the system rule that no partial credit is awarded.

The risk parameters are placed by the same formula chain. The stop loss buffer scales with confidence, the risk distance is held to a minimum of two times ATR, and the take profit is set at the tier multiplied by the risk distance and capped at six times ATR. With the bar currently loaded on the Input sheet the engine returns TAKE for a BUY at a confidence of 1.0, produced by a displacement ratio of about 4.5 that clamps to the maximum, with all eight gates passing, an entry of 1.0847, a stop at 1.07835 and a target at 1.09554. This demonstrates the complete pipeline producing a live and internally consistent result that the screen recording walks through end to end.

3.4 Implementation and Analysis of the Trader Dashboard

Development process and features

The Trader_Dashboard was built to answer a single question for the primary end user in seconds, which is whether to take the current setup. The development process began by placing the engine decision as a large headline cell, then adding the six parameters a trader acts on, being direction, confidence, risk to reward, entry, stop and target, as linked cells beside it. A recent signals table was then added beneath, drawing the last ten rows from Trade_History, and two charts were placed below that, a column chart of recent profit and loss and a line chart of the cumulative equity curve.

Implementation techniques

The implementation centres on linked cells and conditional formatting. The headline reads the engine result directly so the dashboard and the engine can never disagree, and a conditional formatting rule fills it green for TAKE and red for REJECT so the answer is legible before any number is read. The recent signals table uses conditional formatting to shade wins green and losses red and to colour each profit and loss value by its sign. The column chart is configured to plot trade outcomes as discrete bars and the line chart to plot the running pip total as a continuous series, which are the chart types justified for this view in Stage 2.

Communication of trends, patterns and insights

This view communicates an insight, a trend and a pattern at once. The insight is the take or reject decision carried by the headline, the trend is the slope of the rising equity curve, and the pattern is the recent run of wins and losses in the table. The visual hierarchy places the single most important output at the top left where the eye lands first, which is exactly the behaviour the time pressured trader persona needs, and the effect is a dashboard that delivers a confident decision and the recent context for it without the trader having to interpret a single raw calculation.

Screenshot

3.5 Implementation and Analysis of the Coach Dashboard

Development process and features

The Coach_Dashboard was built to expose the reasoning behind a decision so a mentor can teach it, which is the opposite priority to the trader view. The development process began with a price context strip at the top, then the centrepiece, an eight row evaluation table that names each Module E filter alongside its condition, its tested value, its PASS or FAIL result and a plain language explanation of what the filter protects against. The overall decision was placed beneath the table with a hyperlink through to the Calculation_Breakdown sheet, and a scatter plot of displacement ratio against trade profit and loss was added last to show the relationship between setup strength and outcome.

Implementation techniques

Every tested value and result in the table is a linked cell that reads the live gate output from the Logic sheet, so the coach view always shows the same evaluation the engine performed rather than a retyped copy. Conditional formatting shades each result cell green for PASS and red for FAIL, and the plain language explanation column carries the meaning in words so the table never relies on colour alone. The Calculation_Breakdown sheet supports the table by deriving the numbers step by step, from candle body and fourteen bar average body through to displacement, confidence, the risk to reward tier and the placed risk parameters, with each filter gate stated as the native Excel formula condition it tests rather

than as code. The scatter plot is configured to place displacement ratio on the horizontal axis and trade outcome on the vertical axis so the relationship is visible as a cloud of independent points.

Communication of trends, patterns and insights

This view communicates the logic of the system rather than a single answer. The pattern a coach draws out is which combinations of passing filters tend to precede profitable trades, the relationship is the upward drift of profit as displacement strength increases, and the insight is that the value of the system lies in the confluence of all eight gates rather than in any one of them, so the strength of a setup, not merely its existence, is what makes it worth taking. Giving the evaluation table the central visual weight makes the dashboard read as a teaching narrative that moves from observation to reasoning to result.

Screenshot

3.6 Implementation and Analysis of the Institutional Dashboard

Development process and features

The Institutional_Dashboard was built to answer a governance question, which is whether the strategy is preserving capital within risk tolerance. The development process began with a band of headline performance indicators across the top, being total return, win rate, average win against average loss, profit factor, maximum drawdown and average risk to reward tier. A tolerance flag was placed directly beneath them, then a per pair breakdown table for EUR/USD, GBP/USD and USD/JPY, and finally a cumulative return line chart and a horizontal bar chart of net return by currency pair.

Implementation techniques

The headline metrics are produced by aggregation formulas that read the accumulating Trade_History log, using SUM and COUNT for totals, COUNTIF and COUNTIFS for win rates, AVERAGEIF for average win and loss sizes and SUMIF for the profit factor. Maximum drawdown is read from a running peak and drawdown pair of columns maintained within Trade_History, and the tolerance flag is an IF formula that compares that drawdown against an agreed 120 pip band and returns WITHIN TOLERANCE or BREACH, reinforced by conditional formatting. The per pair table reuses the same conditional aggregation functions filtered by currency pair so the manager can see which pairs are carrying the result. With the current log of 37 trades the dashboard reports a total return of 493 pips, a win rate of about 56.8 per cent, an average win of about 39.6 pips against an average loss of about 21.2 pips, a profit factor of about 2.46 and a maximum drawdown of 87.7 pips, which sits within the 120 pip tolerance.

Communication of trends, patterns and insights

This view communicates trend, pattern, relationship and insight at the portfolio level. The trend is the direction of the cumulative return across the period, the pattern is the relative consistency of win rate from one pair to the next, and the relationship is between risk taken, read as maximum drawdown, and reward earned, read as total return. The insight that supports the governance decision is whether the strategy deserves continued or increased capital, and presenting the headline figures as large numeric indicators rather than charts lets that judgement be reached in seconds, with precision preserved for a compliance setting.

Screenshot

3.7 Implementation and Analysis of the Operational Dashboard

Development process and features

The Operational_Dashboard was built to answer the administrator question, which is whether the system is sound so that its results can be trusted. The development process began with a system status table holding one row per component, being historical data feed completeness, evaluation validation, error rate and data freshness, each carrying a metric, a value and a traffic light status. An evaluation cycle log was added beneath to record errors and evaluations across recent cycles, and a compact error rate trend was placed alongside it.

Implementation techniques

Each status row resolves through an IF formula that converts a measured value into an operational, degraded or failed judgement, so the colour and word are computed from the data rather than set by hand. The data feed row counts the bars available in Historical_Data and confirms the dataset meets its minimum, the error rate row divides total failed evaluations by total evaluations across the logged cycles, and the freshness row reports the age of the import. Conditional formatting carries the green, amber and red traffic light convention, and every status is paired with its text label so meaning never rests on colour alone. With the current dataset the data feed reports 150 of 150 bars complete and the error rate resolves to about 1.4 per cent, both of which sit in the operational band.

Communication of trends, patterns and insights

This view communicates system soundness and a reliability trend. The trend is whether the error rate is rising across successive evaluation cycles, the pattern is whether any failures cluster around a particular condition, and the insight is whether the stored dataset and the evaluation process can be trusted. Because the data is static, these insights describe the integrity of the dataset and the engine rather than the liveness of a market feed, and arranging the status indicators as a single scannable column lets one downward glance confirm that every component is sound.

Screenshot

3.8 Explanation of the Automation and Dynamic Recalculation Process

Because Excel cannot subscribe to a live broker feed, the automation in this solution is the manual trigger cycle, in which a single change on the Input sheet recalculates the entire dependency chain at once. When the user enters a new bar, Excel re evaluates the Logic sheet, every filter gate, the displacement and confidence calculations and the placed risk parameters, and then refreshes every dashboard that references them in the same instant. The formula dependency graph is itself the automation, so no macro or external code is required, and the update is transparent because each result can be traced back through the cells that produced it.

This is the appropriate form of automation for a static dataset tool. It guarantees that the trader headline, the coach evaluation table, the institutional aggregates and the operational status all reflect the same input without any manual copying, and it preserves the audit trail that a coded macro would hide. Where Phase 1 relied on live Pine Script execution, Phase 2 reproduces the same logic as a recalculating spreadsheet, which trades real time streaming for full visibility of every step, a deliberate and defensible choice for a decision support and teaching tool.

3.9 Description of the Data Integrity and Security Measures

Data integrity is protected at every stage of the build. The Historical_Data sheet is the single source of truth, so no figure is duplicated and the four dashboards cannot drift apart. Every analytics and dashboard value is formula derived rather than typed, which means a corrected price flows through automatically and no stale number can be left behind. The Input sheet applies data validation so that only sensible entries are accepted, division operations are protected by a non zero floor so an empty cell cannot produce an invalid result, and a decision is only issued once all eight gates have been evaluated, so no signal is ever produced from incomplete data. The completed workbook was recalculated in full and returns zero formula errors across roughly 1790 formulas, which is direct evidence that the integrity measures hold.

Security is handled through separation and protection. The Historical_Data sheet is hidden and treated as read only so the source of truth is out of reach of accidental editing, the input cells are the only

region a user is meant to change, and the engine sheets are kept apart from the dashboards so a stakeholder reviewing a view cannot overwrite the logic behind it. The static nature of the dataset is stated clearly on every dashboard, with the period ending date shown, so the figures can never be misread as live market performance. Together these measures keep the data both trustworthy and protected from inadvertent or misleading change.

3.10 Evaluation of the Effectiveness of the Implemented Visualisations

Judged against the needs of each stakeholder, the implemented dashboards communicate their intended trends, patterns and insights highly effectively. The trader view reduces an eight gate evaluation to one legible verdict and the recent context for it, which is exactly what a time pressured user needs. The coach view decomposes that same verdict into eight inspectable reasons and a relationship plot, which makes the system teachable rather than opaque. The institutional view collapses the full trade history into headline governance metrics with a tolerance judgement attached, and the operational view confirms in a single glance that the data and the engine can be trusted. Each view uses the chart type justified in Stage 2, dual encodes colour with text for accessibility, and draws from one consistent dataset, so the four dashboards form a single coherent system rather than four separate tools.

The main limitation is that the institutional and operational views depend on an accumulating trade history and a static dataset rather than live data, so their figures are representative of the modelled period rather than the current market. This limitation is handled honestly by labelling the static period on every dashboard and by monitoring dataset completeness on the operational view, so the stakeholder is never misled about what the numbers represent. Taken as a whole, the solution comprehensively implements highly effective, stakeholder focused dashboards that communicate the decisions, the reasoning, the performance and the soundness of the Decision Support System, with the live build demonstrated end to end in the accompanying screen recording.

Stage 4:

This stage tests the data visualisation solution against defined, stakeholder focused criteria, records the testing from the perspective of training the stakeholders who will use the Decision Support System, and then evaluates how effectively the visualisations support interpretation and decision making. The evaluation considers filtering, aggregation, depth of insight, clarity of communication, potential bias,

security and overall user experience, and closes with realistic improvements that would enhance clarity, accessibility and stakeholder usefulness. The live testing and training walkthrough are demonstrated in the accompanying screen recording.

4.1 Description of the Testing Approach and Test Criteria

Testing was carried out in three layers so that the solution could be verified from the engine outward to the stakeholder. The first layer is functional testing of the decision engine, which confirms that the underlying logic produces the correct decision, confidence, tier and risk parameters and that it fails safely on edge cases. The second layer is functional testing of the four dashboards, which confirms that each view reports the engine and the trade history faithfully and that its formatting, aggregation and protection behave as designed. The third layer is usability testing from the stakeholder training perspective, which confirms that each stakeholder can reach the decision their dashboard is built for.

The test criteria are stakeholder focused rather than abstract. Each criterion is expressed as the behaviour a stakeholder depends on, for example that the trader headline can never disagree with the engine, that the institutional total can never drift from the trade log, and that a calculated cell cannot be overwritten by accident. A test passes only when the actual result matches the expected result exactly, and the full workbook resolves with zero formula errors across roughly 1790 formulas, which is the baseline condition for every test below.

4.2 Functional Testing of the Decision Engine

The engine was tested against the eight criteria below by loading specific bars on the Input sheet and reading the recalculated result on the Logic sheet. Every test passed.

#	Test criterion	Method	Expected	Actual	Result
E1	All eight gates passing produces TAKE	Load the qualifying bar (close 1.0847)	TAKE, 8 of 8 gates pass	TAKE, 8 of 8	Pass
E2	Any single gate failing produces REJECT	Shrink the candle body so displacement falls below 1.5x	REJECT, gate E5 = FAIL	REJECT, 6 of 8, E5 FAIL	Pass
E3	Confidence is clamped to the 0 to 1 range	Load a bar with a displacement ratio of about 4.5	Confidence = 1.00	1.00	Pass
E4	Risk-reward tier follows the confidence bands	Read the tier at confidence 1.00	Tier = 3:1	3:1	Pass

#	Test criterion	Method	Expected	Actual	Result
E5	Risk distance respects the two times ATR floor	Compare risk distance with 2x ATR on Logic	Risk distance not below 2x ATR	Held at floor	Pass
E6	Take profit is capped at six times ATR	Compare target distance with 6x ATR	Target distance not above 6x ATR	Capped	Pass
E7	Division by zero is handled safely	Set candle high equal to low (range = 0)	Efficiency = 0, no error	0, no error	Pass
E8	Stop sits on the correct side of entry	Inspect the BUY stop against entry	Stop below entry	1.07835 < 1.0847	Pass

4.3 Functional Testing of the Stakeholder Dashboards

Each dashboard was then tested against the behaviour its stakeholder relies on, covering linked values, conditional formatting, aggregation, protection and navigation. Every test passed.

Table

4.4 Testing from the Stakeholder Training Perspective

Beyond confirming that the solution is correct, testing checked that each stakeholder could be trained to use it quickly, because a dashboard that is accurate but unusable does not support a decision. Each view was tested from the position of a stakeholder being shown it for the first time.

The trader needs almost no training, because the headline verdict and the six action parameters carry the whole decision, and a first time user reaches the take or wait choice without an explanation of the underlying calculation. The coach needs a short orientation to the eight filter table and the calculation breakdown sheet, after which the reasoning behind any decision can be traced filter by gfilter, which is the behaviour a mentor must be able to teach. The institutional user is trained to read the headline indicators and the tolerance flag first, and testing confirmed that the governance judgement of whether the strategy sits within tolerance can be made in seconds without reading the trade log. The administrator is trained to read the status column and the cycle log, and testing confirmed that an at a glance soundness check is possible from the traffic light states alone. This training walkthrough is the spine of the screen recording, which shows each stakeholder being guided through their view.

4.5 Evaluation of the Visualisations Against Quality Criteria

Depth of insight

The solution delivers genuine depth rather than a single headline. The trader view surfaces the decision, the recent run of outcomes and the equity trend, the coach view decomposes one verdict into eight inspectable reasons and a setup-quality relationship, and the institutional view turns a trade log

into win rate, profit factor, drawdown and a tolerance judgement. Each stakeholder is given exactly the depth their decision needs and no more, which is the correct calibration for a decision support tool.

Filtering and aggregation

Filtering and aggregation were tested as the mechanisms that shape interpretation. The eight Module E filters are the filtering layer, and testing confirmed that a single failing filter is enough to change the decision from TAKE to REJECT, so the filtering is strict and behaves as a logical AND. The institutional aggregation was tested against the raw trade log and reproduces the totals exactly, with win rate, average win and loss, profit factor and drawdown all derived by formula from the same 37 trade history. The aggregation therefore compresses the data without distorting it, which is what a governance reader depends on.

Clarity of communication

Clarity is high because each dashboard reduces its message to the one decision its stakeholder makes and places the most important output where the eye lands first. The verdict is legible before any number is read, colour and text carry meaning together, and the chart types match the data shape. The main residual clarity risk is that a viewer could read the static figures as live performance, which is mitigated by stating the data period on every dashboard.

Potential bias

Several sources of potential bias were identified and handled honestly. The confidence score reflects displacement strength only and not a probability of profit, so a high score could create an overconfidence bias, which is mitigated by labelling the score for what it measures and by holding position sizing to the risk-reward rule. The per pair aggregates are drawn from a small sample, with some currency pairs carrying only six or seven trades, so a high win rate for those pairs is not statistically robust and should be read as indicative rather than reliable. The static dataset captures a single period and therefore carries period bias, mitigated by labelling the window. Finally the green and red convention would bias against users with colour vision deficiency, which is mitigated by the text labels that accompany every coloured state.

Security

Security was tested directly. Every sheet is protected, the manually entered Input cells are the only editable region, and the hidden Historical_Data sheet is locked, so an attempt to overwrite a calculated or engine cell is rejected while a legitimate new bar can still be entered. This was confirmed by attempting both edits during testing. The static dataset is clearly labelled so the figures cannot be misrepresented as live, which protects the integrity of any decision taken from them.

Overall user experience

The overall user experience is strong because the four dashboards form one consistent system. The colour convention, column order and layout are shared across views, navigation between them is a

single click, and each view is reduced to a scannable screen rather than a dense grid. A user who learns one dashboard can read the others without relearning, which lowers the training cost and supports confident use under time pressure.

4.6 Justification of Realistic Improvements

Several realistic improvements would raise clarity, accessibility and stakeholder usefulness further. Replacing the manual paste import with a Power Query connection would let the dataset refresh from a saved export at the press of a button, which would reduce the risk of stale data and make the tool easier for a non technical stakeholder to keep current. Adding a visible data as at stamp with an automatic warning when the import date exceeds a chosen age would further guard against reading static figures as live. For accessibility, supplementing the green and red convention with shapes or patterns, and offering a colour blind safe palette, would remove the last dependence on hue. For training and clarity, attaching a short cell comment to each headline indicator would let a new stakeholder learn what a figure means in context rather than from a manual. Finally, widening the trade history and adding more currency pairs would make the per pair aggregates statistically robust and reduce the small sample bias noted above. Each of these is achievable within Excel and is justified by a specific weakness identified during testing rather than added for its own sake.

4.7 Evaluation of Support for Stakeholder Decision Making

Taken together, the testing evidence shows that the solution supports each stakeholder decision it was built for. The trader reaches a take or wait decision in seconds, the coach can teach the reasoning behind it, the fund manager can judge whether the strategy sits within tolerance, and the administrator can confirm the system is sound. The decision engine fails safely, the aggregation is faithful, the security holds, and the residual limitations are identified honestly and matched to realistic improvements. The solution therefore comprehensively meets its purpose as a stakeholder focused decision support tool, with the testing and training demonstrated end to end in the accompanying screen recording.

Feedback:

Go more in detail for example which place is the data coming from for the chart generation where is that getting it from and highlight syllabus words words like clarity and what not to make it easier for her to mark.also annotated screenshots ... the videos are really important